NYANZA ADVENTIST SECONDARY SCHOOL
PRE-MOCK 1 FORM FOUR

041 BASIC MATHEMATICS

TIME: 2:30 Hrs **Year: 2025**

INSTRUCTIONS

1. This paper consists of sections A and B with a total of fourteen questions.
2. Answer all questions in both sections.
3. Each question in section A carries six (06) marks while each question in section B carries ten (10) marks.
4. Non programmable calculators may be used.
5. Cellular phones and all unauthorized materials are not allowed in the examination room.
6. Where necessary use R= 6370km and $\pi = 22/7$.

SECTION A (60 Marks)
Answer all questions in this section.

1. a) The government provided tree seedlings to 895 villages for reforestation. If each village got 96857 tree seedlings, how many tree seedlings did the government distribute to all villages?

   b) A boy spent 35% of his money to buy books. If he was left with 26,000 Tanzania shillings, how much money did he have?

2. a) The volume (V) of a cylinder with a base of radius r and height h is given by $V = \pi r^2 h$. Make r the subject of the formula

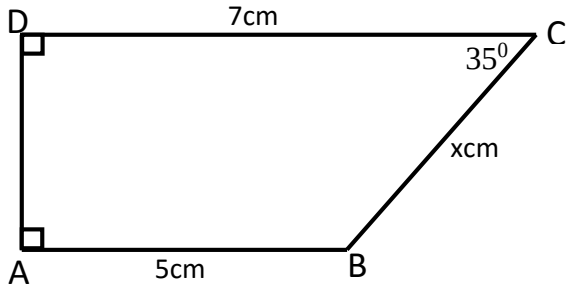
   b) Given that, log 2 = 0.3010 and log 3 = 0.4771. Calculate the value of log 48 and write the laws involved at the stage of finding the answer.

3. a) In a certain street with 200 houses, 170 houses have electricity and 145 houses have glass doors. How many houses have both electricity and glass doors? Assuming that each house has either a glass door or electricity or both.

   b) A piece of chalk is picked at random from a box containing 5 identical pieces, two of which are red and three are white. Find the probability that the piece of chalk picked is red.

4. a) Find the equation of the straight line in the form of $y = mx + c$ passing through the point (5,3) and (6,7)
   b) The forces of magnitudes 5N and 7N act on a particle at an angle of 90° between them. Find the magnitude and direction of their resultant.

5. a) The following figure shows a trapezium ABCD. Calculate
      i) the value of x
      ii) the area of the trapezium ABCD



   b) Given that ΔPQR ~ Δ LMN and ΔPQR ~ ΔABC, identify the corresponding sides and corresponding angles Δ ABC and Δ LMN.

6. a) Three pieces of a wire measure 96 decameters, 3 metres and 130 millimeetres respectively. Find the total length of the three pieces in kilometers.

   b) A man had two different plants. He measured the rate at which the plants were growing. Plant A grew 5cm in 10 days and plant B grew 8cm in 12 days. Which plant was growing more quickly?

7. a) A house was sold at a profit of 1,900,000 Tanzania shillings. If the percentage profit was 25%. What was the buying price of the house?

   b) On 1st July 2023, Juma started a business with the capital in cash of 1,000,000 and made the following transactions:

July2	Purchased goods for cash	Tsh.700,000
July3	Sold goods for cash	Tsh.500,000
July 6	Purchased goods for cash	Tsh.300,000
July 10	Bought goods for cash	Tsh.235,000
July15	Paid cash for rent	Tsh.110,000
July 26	Paid cash for wages	Tsh.55,000
July 28	Sold goods for cash	Tsh.310,000

   Use these transactions to prepare cash account and balance it.

8. a) Ali saved Tsh.2, 000,000 in the first year of his employment and each year afterwards he saved Tsh.400, 000 more than the previous year. How much money will he have saved in the first 22 years?

   b) If the sum of the first four terms of a geometric progression is 80 and its common ratio is 3. Find the first term.

9. a) The adjacent sides of a right-angled triangle have lengths 5cm and 12cm. Find the length of the hypotenuse.

   b) The angles of elevation of a balloon from two points A and B which are 0.3km apart are 62°and 48° respectively. If the balloon is vertically above the line AB, find its distance above the line.

10. a) The lengths of the parallel sides of a trapezium are (x + 4) and (x + 2). The perpendicular distance between the parallel sides is (2x - 3). Find the area of the trapezium and write your answer in the form of quadratic expression.
    b) John bought a certain number of pens for 3600 shillings. If each pen had costed 50 shillings less, he could have bought six more pens for the same amount of money, how many pens did he buy?

SECTION B (40 Marks).

Answer all questions in this section.

11. a) The height of players was recorded in centimeters as shown in the table below.

Height (cm)	120 -125	124 -129	130 -134	135 - 139	140 -144	145 -149	150 -154
Frequency	3	2	4	12	7	6	2

   i) Calculate the Mean height and Median height
   ii) Use the mean above to interpret the data

   b) Prove that, the angle at the Centre of the circle is twice the angle at the circumference given that they are formed by the same arc.

12. a) An aeroplane flies from Tabora (5°S, 33°E) to Kilimanjaro (5°S, 39°E) at 322 Kilometres per hour along a parallel of latitude. If it leaves Tabora at 3:00 pm, find the arrival time at Kilimanjaro airport.

b) The lengths of two sides of a triangle are 10 cm and 12 cm. If the area of a triangle is $58cm^2$, find the value of the included angle.

13. a) Given A= $\begin{pmatrix} 4 & 4 \\ 2 & 1 \end{pmatrix}, B = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} -2 & 4 \\ 4 & 5 \end{pmatrix}$; find $A(B + C)$ and $(AB)^2$

b) Peter is running a business within a Samia schools' compound; he is selling pieces of cake and soda. Each piece of cake costs 1500 shillings and each bottle of soda costs 500 shillings. In one afternoon, he made a total of 255,500 shillings by selling a total of 225 pieces of cake and soda. Use the given information to:

i. Formulate the system of simultaneous equations.
ii. Transform the system of linear simultaneous equations in a matrix form
iii. Use the inverse method of matrix to find number of pieces of cake and bottles of soda sold.

14. a) A business woman has 120,000 shillings to spend on tomatoes. At market A a tomato costs 800 shillings and at market B a tomato costs 1200 shillings. Market A has only 60 tomatoes left, and a business woman wants to obtain the greatest number of tomatoes using the money she has;

(i) Formulate a linear programming problem
(ii) Find the greatest number of tomatoes a business woman can buy.

b) Given $R = \{(x, y): y = x^2 - 4\}$; find domain and range of the inverse of the given relation R.